

A Central Finite-Trace Case of Semifinite Liftability

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Status. This packet records a likely valid partial result for the question in Y. Ueda, *On the predual of non-commutative H^∞* , arXiv:1002.3672:

`partial_result_likely_valid.`

Source question. Ueda proves that for a finite tracial non-commutative H^∞ -algebra $A = H^\infty(M, \tau)$, every weakly relatively compact subset of $M_*/A_\perp$ has weakly relatively compact unique normal Hahn–Banach extensions in M_* . In Section 3.1, after proving the basic semifinite prerequisites, the paper asks whether this liftability survives in the semifinite tracial setting.

Partial answer. The answer is positive when the semifinite algebra is a countable central sum of finite-trace pieces. This includes precisely the case where the semifinite difficulty is only an ℓ^∞ -direct-sum bookkeeping issue, not a noncentral or continuous exhaustion issue.

Proof Intuition. Central finite projections split both the von Neumann algebra and the subdiagonal algebra into independent finite summands. On each summand, Ueda’s finite theorem gives the desired liftability. The only remaining point is to assemble the pieces. Weak compactness in an ℓ^1 -sum is controlled by two conditions: coordinatewise weak compactness and uniformly small ℓ^1 -tails. The coordinate extension maps preserve norms, so the tail condition is unchanged by lifting.

Lemma 1 (ℓ^1 -sum weak compactness criterion). *Let X_n be Banach spaces and $X = (\oplus_{n=1}^\infty X_n)_1$. A bounded set $K \subset X$ is relatively weakly compact if and only if:*

1. *for every n , the coordinate set $\pi_n K$ is relatively weakly compact in X_n ;*

2. $\lim_{N \rightarrow \infty} \sup_{x \in K} \sum_{n > N} \|x_n\| = 0$.

Proof. The coordinate condition follows from continuity of π_n . For the tail condition, if it failed, choose $x^{(N)} \in K$ with tail after N at least $\varepsilon > 0$, and choose norm-one functionals supported on that tail which almost attain the tail norm. These functionals are weak-star null in $X^* = (\oplus_n X_n^*)_\infty$, contradicting uniform convergence on weakly compact subsets.

Conversely, take any sequence in K . By the coordinate weak compactness and a diagonal extraction, pass to a subsequence whose every coordinate converges weakly. The uniform tail condition then shows that the coordinate limits form an element of X , and testing first on finitely many coordinates and then using the small tails gives weak convergence in X . The Eberlein–Smulian theorem completes the proof. $\square$

Theorem 1 (Central finite-trace liftability). *Let M be a sigma-finite von Neumann algebra with faithful normal semifinite trace Tr , and let*

$$A = H^\infty(M, \text{Tr})$$

be a semifinite tracial non-commutative H^∞ -algebra. Suppose that there are mutually orthogonal central projections $z_n \in D = A \cap A^$ such that $\sum_n z_n = 1$ strongly and $0 < \text{Tr}(z_n) < \infty$ for all nonzero z_n . Then every weakly relatively compact subset*

$$W \subset M_*/A_\perp$$

has weakly relatively compact unique normal Hahn–Banach extensions in M_ .*

Proof. Set $M_n = z_n M$, $A_n = z_n A$, and normalize the finite trace on M_n by

$$\tau_n(x) = \text{Tr}(z_n)^{-1} \text{Tr}(x), \quad x \in M_n.$$

Since z_n is central and lies in D , the conditional expectation for A restricts to the conditional expectation for A_n , and each

$$A_n = H^\infty(M_n, \tau_n)$$

is a finite tracial non-commutative H^∞ -algebra.

The central decomposition gives

$$M = \bigoplus_{n=1}^{\infty} \ell^\infty M_n, \quad M_* = \bigoplus_{n=1}^{\infty} \ell^1 (M_n)_*,$$

and similarly

$$A = \bigoplus_{n=1}^{\infty} \ell^\infty A_n, \quad A_\perp = \bigoplus_{n=1}^{\infty} \ell^1 (A_n)_\perp.$$

Hence the quotient isometrically identifies with

$$M_*/A_\perp \cong \bigoplus_{n=1}^{\infty} \ell^1 ((M_n)_*/(A_n)_\perp).$$

Indeed, the quotient norm separates coordinatewise:

$$\inf_{\eta \in A_\perp} \sum_n \|\rho_n + \eta_n\| = \sum_n \inf_{\eta_n \in (A_n)_\perp} \|\rho_n + \eta_n\|.$$

Let $X_n = (M_n)_*/(A_n)_\perp$ and $Y_n = (M_n)_*$. For each n , Ueda's finite theorem says that the unique normal Hahn–Banach extension map

$$S_n : X_n \rightarrow Y_n$$

sends relatively weakly compact subsets of X_n to relatively weakly compact subsets of Y_n . It is also isometric:

$$\|S_n x_n\| = \|x_n\|_{X_n}.$$

Let $W \subset (\oplus_n X_n)_1$ be relatively weakly compact. By the lemma, each coordinate set $\pi_n W$ is relatively weakly compact and

$$\lim_{N \rightarrow \infty} \sup_{x \in W} \sum_{n > N} \|x_n\| = 0.$$

Therefore $S_n(\pi_n W)$ is relatively weakly compact in Y_n for every n , and the lifted set

$$S(W) = \{(S_n x_n)_n : x = (x_n)_n \in W\} \subset (\oplus_n Y_n)_1$$

has the same tails:

$$\sum_{n>N} \|S_n x_n\| = \sum_{n>N} \|x_n\|.$$

Applying the lemma again in the ℓ^1 -sum of the Y_n 's shows that $S(W)$ is relatively weakly compact in M_* .

Finally, $S(W)$ is exactly the set of unique normal Hahn–Banach extensions to M . Coordinate-wise uniqueness gives uniqueness on the ℓ^∞ -sum, and the norm equality above gives the required norm-preserving extension. This proves the claimed liftability. $\square$

Verification Notes. The proof uses only Ueda's finite liftability theorem, the central direct-sum identification, and the standard weak compactness criterion for ℓ^1 sums. No computation is involved. The most important point for human review is the quotient identification

$$M_*/A_\perp \cong \bigoplus_1 (M_n)_*/(A_n)_\perp,$$

which uses centrality of the projections z_n .

Limitations. This is not a full solution to Ueda's semifinite question. The argument does not cover increasing noncentral finite projections $p_m \in D$, continuous central decompositions, semifinite factors such as $B(\ell^2)$ with its usual trace unless $A = M$ trivially, or not-necessarily-tracial non-commutative H^∞ -algebras.

Novelty Check. Before promotion, the run checked the lightweight indexes for arXiv:1002.3672 and the phrases “predual of non-commutative H^∞ ”, “weakly relatively compact”, “semifinite”, and “liftability”. A web search on the exact liftability and semifinite phrases found the source paper and metadata pages, but no later exact resolution of the semifinite liftability question. This packet should therefore be treated as a narrow partial theorem, not as a known full literature answer.

Human Review Recommendation. Likely valid partial result. Review should focus on the direct-sum quotient identification and on whether the central-decomposition hypothesis is stated in the cleanest natural form.

References.

- Y. Ueda, *On the predual of non-commutative H^∞* , arXiv:1002.3672; Bull. London Math. Soc. 43 (2011), 886–896.